AS · Unit 1 · WBI11 Molecules, Diet, Transport and Health

Membranes and transport

Every cell is enclosed by a membrane that decides what gets in and what gets out. This topic explains how the membrane is built from phospholipids and proteins, and how substances cross it by passive movement, by using energy, or by being packed into vesicles.

SPEC 2.1 Gas exchange surfaces, Fick's law and the lung

WHAT THIS SPEC POINT ASKS

(i) Know the properties of gas exchange surfaces: large surface area to volume ratio, thin surface, concentration difference. (ii) Understand how these set the rate of diffusion, and calculate it with Fick's law. (iii) Understand how the mammalian lung is adapted for rapid gas exchange.

The three properties

- **Large surface area** (relative to volume): more area for molecules to cross at once.
- **Thin surface / short diffusion distance**: molecules have less far to travel.
- **Steep concentration gradient**: kept steep by bringing in fresh supply on one side and carrying away on the other.

Big organisms have a **larger surface area** but a **smaller SA:V ratio**, so they need specialised exchange surfaces (lungs, gills, leaves) and a transport system (1.6).

Fick's law

rate of diffusion ∝ (surface area × difference in concentration) ÷ thickness of surface

- Double the area → double the rate. Double the concentration difference → double the rate. **Double the thickness → halve the rate.**

WORKED EXAMPLE

Worked example: Fick's law

Alveolus A: area 70 units, concentration difference 10 units, thickness 0.5 μm → rate ∝ (70 × 10) ÷ 0.5 = **1400**.

Alveolus B after thickening (e.g. fibrosis): thickness 1.0 μm → (70 × 10) ÷ 1.0 = **700**.

Doubling the thickness halved the rate. Questions often ask for the **percentage change**: (700 − 1400) ÷ 1400 × 100 = **−50%**.

How the lung is adapted

Feature	How it speeds up gas exchange
Millions of **alveoli** (and branching bronchioles)	**huge surface area** (about 70 m² in humans)
Alveolar wall is **one cell thick** of **flattened (squamous) epithelium**	**short diffusion distance**
Capillary wall is one cell thick, pressed against the alveolus; red cells squeezed against the wall	short distance (total barrier about 0.5 μm)
Dense **capillary network** around each alveolus; blood constantly flowing	removes O_2 and brings CO_2 → **maintains the concentration gradient**
Ventilation (breathing)	replaces alveolar air → keeps O_2 high and CO_2 low in the alveoli → maintains the gradient
Moist lining	gases dissolve before diffusing across
Elastic fibres	stretch on breathing in, recoil to push air out
Surfactant	stops alveoli collapsing (beyond spec)

Airways: the **trachea** and **bronchi** are held open by **cartilage**; ciliated epithelium and **goblet cells** (mucus) trap and remove particles; **bronchioles** have smooth muscle to change their diameter.

MARK-SCHEME WORDING

Fick's law: "rate ∝ (surface area × difference in concentration) ÷ thickness of surface".

Link every feature to rate: "alveoli → **large surface area** → faster diffusion" · "one cell thick / squamous epithelium → **short diffusion distance**" · gradient "**maintained by ventilation / blood flow**". A list of features with no links scores 1.

Disease (emphysema, CF): "smaller surface area / longer distance / lower gradient" → "diffusion of oxygen is **slower**". **Ignored**: "less gas exchange", "less efficient". **Not accepted**: "no diffusion".

Factors other than distance: surface area, concentration gradient, permeability, temperature. "SA:V" is **not accepted** here.

Often asked as: name the law; predict the effect of halving or doubling one term; explain lung adaptations; explain how a disease or an unfamiliar surface (egg shell, amphibian skin) changes the rate.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Temperature** and the **diffusion coefficient** (gases diffuse faster through air than through a membrane) also affect the rate (WBI11 Jan 2021).
- **Behaviour can keep the gradient steep**: a salamander rocks its body to stir the water over its skin, replacing deoxygenated water (WBI11 Oct 2022).
- **Unfamiliar surfaces** (amphibian skin, egg-shell pores): apply the three Fick's-law terms to whatever the question shows you (WBI11 Jan 2021).

COMMON MISTAKE

Saying the alveoli have thin **cell walls**. Animal cells don't have cell walls: say "the **alveolar wall** is one cell thick". Also, don't confuse "large surface area" with "large SA:V ratio".

Calling it Bohr's or Boyle's law. It's **Fick's law**.

Writing "more diffusion" or "more efficient". Say "**faster** diffusion" or "a higher **rate** of diffusion".

EXAM TIP

For "explain how the lung is adapted" answers, give the **feature + which Fick's-law term it affects**: "many alveoli → large surface area → faster diffusion".

SPEC 2.2 Cell membranes and the fluid mosaic model

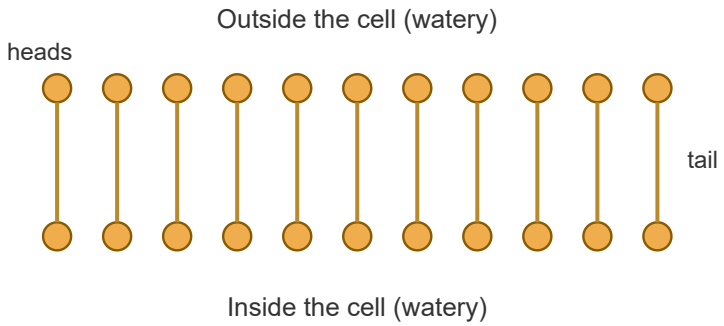
WHAT THIS SPEC POINT ASKS

(i) Know the structure and properties of cell membranes. (ii) Understand how models such as the fluid mosaic model are interpretations of data, used to develop scientific explanations of membrane structure and properties.

The parts of the membrane

- **Phospholipid bilayer**: each phospholipid has a **hydrophilic phosphate head** (faces the water inside and outside the cell) and **two hydrophobic fatty acid tails** (point inwards, away from water). The bilayer forms spontaneously in water.

- **Proteins**:
  - **Intrinsic** (integral) proteins span the bilayer: **channel proteins** and **carrier proteins** (2.5).
  - **Extrinsic** (peripheral) proteins sit on one surface: enzymes, receptors, structural supports.
  - A protein's position depends on where its hydrophobic and hydrophilic R groups are.
- **Cholesterol**: sits between phospholipids, with its small polar **OH group next to the phosphate heads** and the rest of the molecule **among the fatty acid tails**. It **regulates fluidity**: it makes the membrane less fluid at high temperatures and stops it becoming too rigid at low temperatures; adds stability.
- **Glycoproteins and glycolipids**: carbohydrate chains on the outer surface, used for **cell recognition**, as receptors (e.g. for hormones) and in cell adhesion.



Why "fluid mosaic"?

- **Fluid**: phospholipids and many proteins move **sideways** within the layer.
- **Mosaic**: proteins are scattered through the bilayer like tiles in a mosaic.

Properties

- **Partially permeable**: small, non-polar molecules (O_2, CO_2) and water pass straight through the bilayer. Ions and large or polar molecules (glucose, amino acids) need **transport proteins**, because the hydrophobic core repels them.

How the model was built from data (models are interpretations)

Model	Evidence it explained	Why it was replaced
Lipid bilayer (Gorter and Grendel, 1925)	lipid extracted from red blood cell membranes covered **twice** the cells' surface area → two layers	didn't explain how polar substances cross, or the role of proteins
Protein–lipid sandwich (Davson–Danielli, 1930s)	electron micrographs showed two **dark lines** (taken to be protein) with a **light band** (lipid) between	membrane proteins turned out to be varied and often hydrophobic; a protein coat would block lipid-soluble molecules

Model	Evidence it explained	Why it was replaced
Fluid mosaic (Singer–Nicolson, 1972)	**freeze-fracture** micrographs showed proteins **embedded in and spanning** the bilayer; labelled proteins from two fused cells **mixed** quickly, so they move	still accepted, refined by later data

The point: scientists propose a model that fits the data; **new evidence** (often from new technology such as electron microscopy) can lead to the model being **modified or replaced**. A model is a representation, not the real thing.

EXAM TIP

Say what **evidence** supported or overturned each model: that's where most marks are. The names have also been examined (a whole question on the history in Jan 2025), so learn them in order (see the box below).

MARK-SCHEME WORDING

Bilayer: "hydrophilic / polar **heads** interact with water" on **both sides**; "hydrophobic / non-polar **tails** are repelled by water", so they point inwards. Describing the arrangement without the reason scores 1.

Drawing a phospholipid: **glycerol + two fatty acids + phosphate**, with every bond line touching.

Cholesterol position: "the polar (OH) region interacts with the **phosphate heads** and the non-polar region is **amongst the lipid tails**".

Fluidity: a higher temperature "increases the **movement of phospholipids**" because "interactions between fatty acid chains break"; cholesterol "**fills the gaps** between the phospholipid tails", so it "decreases fluidity". **Not accepted**: "flexible", "strong".

"Fluid" = phospholipids move; "mosaic" = proteins scattered through the bilayer. "The bilayer moves" isn't credited.

Often asked as: draw or label a phospholipid; explain the bilayer arrangement; read fluidity or permeability graphs with and without cholesterol; compare Davson–Danielli with the fluid mosaic model.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **History of the model, with names** (WBI11 Jan 2025): Overton (lipid-soluble substances enter cells faster, so membranes contain lipid) → Langmuir (lipid monolayer) → Gorter and Grendel (bilayer) → Davson–Danielli (protein–lipid sandwich) → Robertson (unit membrane) → Singer and Nicolson (fluid mosaic, 1972).
- **Cholesterol as a fluidity buffer** (WBI11 Jun 2025): with cholesterol, fluidity rises steadily (roughly linearly) with temperature; without it, the curve is S-shaped with a sharp rise. Cholesterol **raises** fluidity

at low temperatures and **lowers** it at high temperatures.

- **Cold-adapted membranes** (fish in cold water): more **unsaturated** and **shorter** fatty acid chains pack less tightly, so the membrane stays fluid. Fluidity is needed for endocytosis, exocytosis, cell shape changes and cell fusion (WBI11 Jun 2022).
- **Checking a micrograph against the model** (WBI11 Jan 2021): the bilayer is about **5 nm** thick. Actual size = image size ÷ magnification, in the same units: a membrane 0.5 mm thick on a ×100 000 micrograph is 0.5 ÷ 100 000 = 5×10^{-6} mm = **5 nm**.

COMMON MISTAKE

Drawing **three** fatty acids on a phospholipid (that's a triglyceride), or joining the fatty acids to the phosphate.

Saying the heads face water without saying **both sides** of the membrane.

"The phospholipids melt" or "the bilayer moves". Say the **phospholipids move more**.

Describing cholesterol's **function** when the question asks for its **position**.

For cold-water fish, stopping at "keeps the membrane fluid" without the fatty-acid reason, or saying it "keeps the fish warm".

SPEC 2.3 Core Practical 3: membrane permeability

WHAT THIS SPEC POINT ASKS

Investigate membrane properties, including how alcohol concentration and temperature affect membrane permeability.

Principle

Beetroot cells contain a red pigment (betalain) in their vacuoles. When membranes (tonoplast and cell surface membrane) are damaged or more permeable, pigment **leaks out** into the surrounding water. More pigment → darker red → higher **absorbance** in a colorimeter.

CORE PRACTICAL

Core practical 3: method in brief

1. Cut beetroot cores with a **cork borer**, then slice into pieces of **equal size** (same length and diameter, so the same surface area) with a scalpel on a tile.
2. **Rinse** the pieces in distilled water to remove pigment released from cells cut open, then blot dry.
3. **Temperature**: put each piece in the same volume (e.g. 5 cm^3) of distilled water in tubes held in water baths at a range of temperatures (e.g. 0, 20, 40, 60, 80 °C) for the same time (e.g. 30 minutes). Let the water reach temperature first.

4. **Alcohol**: put pieces in the same volume of different ethanol concentrations (e.g. 0, 10, 20, 30, 40, 50%) at the **same temperature** (room temperature) for the same time.
5. Remove the beetroot, shake the tube, and transfer some of the liquid to a cuvette.
6. Zero the colorimeter with distilled water (or the alcohol solution) using a **blue-green filter**, then read the **absorbance** of each sample.
7. Repeat at least three times at each value and calculate means.

Explaining the results

- **Temperature rises** → phospholipids gain kinetic energy and move more, so the bilayer becomes more fluid and gaps appear; above about 40–50 °C **membrane proteins denature**, leaving holes → much more pigment leaks out. (Freezing can also raise permeability: ice crystals pierce membranes.)
- **Alcohol concentration rises** → ethanol is a non-polar solvent that **dissolves the lipids** in membranes and disrupts the bilayer → more pigment leaks.

Limitations and fixes

- Pieces not identical → measure carefully; use the same cork borer.
- Pigment varies across the root → take all pieces from the same root and average.
- Colorimeter cuvettes may differ or be scratched → use the same cuvette, rinsed.
- Time in water baths varies → start the timer as each piece goes in.

MARK-SCHEME WORDING

"same surface area/size of beetroot pieces", "rinse to remove pigment from cut cells", "same volume of water", "same time", "colorimeter / absorbance", "proteins denature", "phospholipids move more / bilayer more fluid", "alcohol dissolves (phospho)lipids"

Temperature mechanism: more kinetic energy → "interactions between fatty acid chains break" → phospholipids move more → more permeable.

Data with two variables (temperature and cholesterol): permeability rises with temperature and falls as cholesterol rises, and "cholesterol has a **greater effect at higher temperatures**". That last point is the one most often missed.

Often asked as: the beetroot method itself wasn't set in WBI11 papers from 2019 to 2026, but permeability and fluidity data questions were, so learn the explanations as well as the method.

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Liposome volume (WBI11 Jan 2019): $V = 4/3\pi r^3$ using the **radius**. A liposome 100 nm across has r = 50 nm, so $V = 4/3 \times \pi \times 50^3 = \mathbf{5.2 \times 10^5\ nm^3}$.

COMMON MISTAKE

Using the **diameter** instead of the radius in sphere formulas: the commonest maths slip.

With two variables, describe each effect **and** how one changes the effect of the other (the interaction).

SPEC 2.4 Osmosis and water potential

WHAT THIS SPEC POINT ASKS

Understand osmosis as the net movement of free water molecules through a partially permeable membrane, down a water potential gradient.

Definitions

- **Osmosis** is the **net movement of free water molecules** from a region of **higher water potential** to a region of **lower water potential**, through a **partially permeable membrane**.
- **Free water molecules**: when a solute dissolves, water molecules cluster around it and are **no longer free** to move across the membrane. More solute → fewer free water molecules → lower water potential.
- **Water potential (ψ)** is measured in **kPa**. **Pure water has the highest possible water potential: 0 kPa.** Any solution has a **negative** value; the more concentrated, the more negative.

WORKED EXAMPLE

Which way does water move?

Cell A: ψ = −400 kPa. Cell B: ψ = −700 kPa. Water moves **from A to B** (from −400 to the more negative −700), i.e. down the water potential gradient.

What happens to cells

Surrounding solution compared with the cell	**Animal cell** (e.g. red blood cell)	**Plant cell**
Higher ψ (more dilute; e.g. pure water)	water enters → swells → may **burst (lysis)**	water enters → vacuole swells → cell becomes **turgid**; the cell wall stops it bursting

Surrounding solution compared with the cell	**Animal cell** (e.g. red blood cell)	**Plant cell**
Same ψ	no net movement	no net movement; cell **flaccid** if already low on water
Lower ψ (more concentrated)	water leaves → cell **shrinks** (crenation)	water leaves → vacuole shrinks → membrane pulls away from the wall: **plasmolysis**

Predicting what happens to a plant cell in a solution

1. Identify the water potential of the solution compared with the cell. A cell's contents usually have a lower water potential than pure water.
2. If the solution is more dilute than the cell (higher water potential), water moves in by osmosis, the vacuole fills, and the cell becomes turgid against the wall.
3. If the solution is more concentrated than the cell (lower water potential), water moves out, the vacuole shrinks, and the cytoplasm pulls away from the wall. This is plasmolysis.
4. Explain each step by pointing to the direction of the water potential gradient across the partially permeable membrane.

CORE PRACTICAL

Recommended practical: tissue water potential

1. Cut equal-sized potato (or other plant tissue) cylinders, blot and **weigh** each.
2. Place each in a different sucrose concentration (e.g. 0, 0.2, 0.4, 0.6, 0.8, 1.0 mol dm^{-3}) for the same time (e.g. 24 h) at the same temperature.
3. Remove, **blot dry** (to remove surface solution) and reweigh.
4. Calculate **% change in mass** for each.
5. Plot % change in mass against concentration. Where the line **crosses zero** (no net gain or loss) the solution has the **same water potential as the tissue**.

WORKED EXAMPLE

Worked example: % change in mass

Start mass 2.50 g, end mass 2.20 g. % change = (2.20 − 2.50) ÷ 2.50 × 100 = **−12%** (a loss: the solution had a lower ψ than the tissue).

Use **% change**, not mass change, because the starting masses differ.

MARK-SCHEME WORDING

Always **compare**: "pond water has a **higher water potential than** the cytoplasm" → "so water moves in **by osmosis**". Hypotonic / hypertonic are accepted only as comparisons.

Animal cell in water: "**no cell wall**" → "can't stop water entering" → "swells / pressure increases" → "**bursts**".

Explaining the definition part by part: "free" = some water molecules are held by solute molecules; "partially permeable" = lets some molecules through but not others; "down a water potential gradient" = from higher to lower ψ.

The **total** concentration of all the solutes (all the ions) sets the water potential.

Ignored: "water concentration", "isotonic". **Wrong**: "the solution moves in".

Often asked as: explain water movement for a named organism or tissue; explain each part of the osmosis definition; 6-mark data questions (contractile vacuoles, crabs moving into sea water, osmotic dehydration of fruit).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Contractile vacuole** (Paramecium in pond water): water keeps entering by osmosis; the vacuole collects it and **pumps it out using ATP**. It doesn't stop water getting in (WBI11 Jan 2020).
- **Osmoconformers** keep their internal water potential the same as sea water, so there's no net osmosis, no shrinking or bursting, and no ATP spent (WBI11 Jun 2023).
- Animals moving into sea water can lower the water potential of their cells with **amino acids** from hydrolysed muscle protein (WBI11 Jan 2022).

COMMON MISTAKE

Writing "water moves from a high to a low **concentration**" or "down a concentration gradient". Use **water potential** every time. Don't say the solute moves: it's the water that moves.

Saying "hypotonic" without saying **compared with what**. Not naming **osmosis** when you explain the water movement.

Asked to explain the definition? Explain each part; don't just write the definition out again.

SPEC 2.5 Passive and active transport, endocytosis and

exocytosis

WHAT THIS SPEC POINT ASKS

(i) Understand passive transport (diffusion, facilitated diffusion), active transport (including ATP as the immediate source of energy), endocytosis and exocytosis. (ii) Understand the roles of carrier and channel proteins in membrane transport.

Summary table

	Diffusion	Facilitated diffusion	Active transport	Endocytosis / exocytosis
Direction	down a concentration gradient	down a concentration gradient	**against** a concentration gradient	bulk movement (either direction)
Energy from ATP?	no (passive)	no (passive)	**yes**	**yes**
Protein needed?	no	**channel** or **carrier**	**carrier** only	no (membrane vesicles)
What moves	small, non-polar (O_2, CO_2); water	ions, polar molecules (glucose, amino acids)	ions, glucose, amino acids	large molecules, particles, whole cells
Example	O_2 into alveolar blood	glucose into red blood cells; Cl^- through CFTR channels	mineral ions into root hairs; glucose reabsorbed in kidney	phagocytosis of bacteria; secretion of enzymes or hormones

The proteins

- **Channel proteins**: water-filled pores lined with hydrophilic R groups. **Ions** and small polar molecules diffuse through. Each is specific; many can open and close ("gated").
- **Carrier proteins**: the molecule binds to a specific **binding site**, the protein **changes shape**, and the molecule is released on the other side. In **facilitated diffusion** this happens down the gradient. In **active transport** the shape change is driven by **ATP hydrolysis**.

Active transport and ATP

1. The molecule or ion binds to the carrier protein on one side.
2. **ATP is hydrolysed** (ATP → ADP + P_i), releasing energy (and the phosphate attaches to the carrier).
3. The carrier changes shape and releases the molecule on the other side, **against** its concentration gradient.
4. The phosphate is released and the carrier returns to its original shape.

ATP is the *immediate* source of energy because its hydrolysis releases energy in **small, manageable amounts**, in **one step**, exactly where it's needed. ATP is made by respiration, so anything that stops respiration (cyanide, no O_2, low temperature) stops active transport.

Bulk transport

- **Endocytosis**: the cell surface membrane **folds inwards** around a substance and pinches off as a **vesicle** inside the cell (e.g. phagocytes engulfing bacteria). Needs ATP.
- **Exocytosis**: **vesicles** (often from the Golgi apparatus) move to the cell surface, **fuse** with the membrane and release their contents outside (e.g. digestive enzymes, insulin, mucus). Needs ATP.

EXAM TIP

Down a gradient with a protein = **facilitated diffusion**. Active transport uses **carrier proteins**, never channels. To show a process is active, look for: movement **against** a gradient, or the rate falling when **respiration is inhibited**.

MARK-SCHEME WORDING

Active transport: "**against** the concentration gradient", "needs **ATP** and a **carrier** protein". State the direction.

Movement through a channel is "**facilitated diffusion**", not "diffusion".

Only **water** moves by osmosis; osmosis moves water, not solute.

Which molecule crosses how: water is "**small** enough" to cross the bilayer (or uses aquaporins); non-polar molecules (O_2, steroids) "diffuse **through the phospholipid tails**"; glucose is **large and polar**, so it can't pass the hydrophobic tails and uses **carrier proteins**; ions use channels or active transport.

Endocytosis: "the membrane **surrounds** the substance". "Pinocytosis" / "phagocytosis" are ignored as names.

Often asked as: Venn diagrams or tick tables of the processes; cloze passages; explain an uptake curve; 6-mark "how do these four molecules cross the membrane".

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- Channel proteins have **polar R groups lining the pore** and **non-polar R groups facing the fatty acid tails** (WBI11 Jan 2024).
- Uptake graphs (WBI11 Oct 2019): diffusion **levels off at equilibrium**; active transport carries on **past** equilibrium, so the inside ends up more concentrated. At a lower temperature, diffusion is slower (less steep) but levels off at the same value.

COMMON MISTAKE

Naming the process without the reason, e.g. "active transport" without "against the gradient, using ATP".

Leaving out that water is **small**, or that non-polar molecules pass **through the tails**.

Sketching a diffusion curve that never levels off.

Quick checks

QUICK CHECK

Which molecule is most likely to cross a phospholipid bilayer by simple diffusion?

- Sodium ions
- Glucose
- Oxygen
- Amino acids

Oxygen is small and non-polar, so it can dissolve in the hydrophobic core of the bilayer. Sodium ions are charged and glucose is a large polar molecule, so both need a protein. Amino acids are polar and cannot pass the core unaided.

QUICK CHECK

A red blood cell is placed in a very dilute salt solution. What is the most likely result?

- It shrinks as water leaves
- It swells and may burst as water enters
- It stays the same size because the solution is isotonic
- It becomes turgid because it has a cell wall

A dilute solution is hypotonic to the cell, so water enters by osmosis. Animal cells have no wall to resist the increase in volume, so they swell and can burst. The turgid answer applies to plant cells, which have a wall.

QUICK CHECK

Which statement best describes active transport?

- Movement down a gradient using a channel protein
- Movement of water through a partially permeable membrane
- Movement against a gradient, using carrier proteins and energy from ATP
- Bulk movement of solid particles through a single pore

Active transport moves substances against the gradient, so it needs ATP and carrier proteins that change shape. Channel-based movement down the gradient is facilitated diffusion, and osmosis concerns water only.

QUICK CHECK

Why does cholesterol help keep a membrane working across a range of temperatures?

- It makes the bilayer fully solid at low temperatures
- It pumps ions across the membrane
- It restrains phospholipid movement when warm and prevents tight packing when cool
- It attaches carbohydrates to the outer surface

Cholesterol stops the membrane becoming too fluid at high temperatures and too rigid at low temperatures. It does not make the membrane solid, pump anything, or attach carbohydrates; those roles belong to other molecules.